User Response Strategy: Annotation Guidelines

1. Purpose

These guidelines are used to annotate how users respond to assistant behavior in multi-turn interactions.

The schema captures whether the user:

- reacts to a problematic assistant response,
- repairs a misunderstanding, insufficiency, or incorrect response,
- steers the assistant's behavior,
- or manages task progression.

The goal is to use user behavior as an observable signal of interaction quality and possible assistant limitations.

This schema can be used for:

- industrial copilots,
- task-oriented assistants,
- spoken dialogue assistants,
- RAG-based assistants,
- multi-turn information-seeking assistants.

Caveat: The schema captures observable user behaviors. Not all follow-up behavior implies conversational breakdown. Some turns reflect normal task progression.

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2. What to Annotate

2.1 Annotation unit

Each annotation unit consists of:

- the current user turn,
- the current assistant response,
- and prior context if needed.

Annotate the current assistant response with:

- `is_trigger`

Annotate the current user turn with:

- `URS_labels`
- `URS_primary`
- `context_target_turn`
- `S_labels`
- `E_signal`
- `rationale`

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3. Core Principle

Annotate **what the user is doing**, not your theory of why the assistant failed.

The user turn is treated as an observable behavioral response to the preceding assistant turn.

Use:

- **Repair** when the user is trying to resolve a problem in understanding, correctness, relevance, or sufficiency of the prior interaction.
- **Steering** when the user is shaping how the assistant should respond.
- **Task Progression** when the user is continuing or managing the task without clear evidence of communicative trouble.

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4. Required Fields

Field	Description
`is_trigger`	TRUE, FALSE, or NA — whether the current assistant turn contains a perceived user-relevant problem
`URS_labels`	All applicable user strategy labels; multi-label allowed
`URS_primary`	One primary URS label
`URS_family`	Derived automatically: Repair, Steering, or Task Management

`context_target_turn`	Prior turn most relevant for interpretation, if not the immediately preceding assistant turn
`S_labels`	Optional stance labels
`E_signal`	Optional end-state signal
`rationale`	Short justification

Notes

- `URS_labels` may be empty.
- `URS_primary` should be assigned only if at least one URS label applies.
- `URS_family` is derived automatically.
- Multi-labeling is allowed, but choose only one `URS_primary`.

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5. Special Cases

5.1 First user turn in a session

If `turn_idx = 1`, annotate only `is_trigger`; do **not** assign URS labels.

5.2 Missing or insufficient context

If the turn cannot be interpreted reliably because context is missing, annotate using available local evidence and mention uncertainty in the rationale.

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6. `is_trigger` — Perceived Assistant Breakdown Signal

Set `is_trigger = TRUE` if the current assistant turn contains a user-relevant problem or insufficiency that could plausibly motivate repair, steering, or dissatisfaction in the immediately following user response.

Typical trigger cases include:

- refusal or inability to help,
- generic fallback without grounding,
- retrieval or document-availability fallback that does not materially help the user progress,

- assistant or tool failure,
- incorrect, misleading, or off-target content,
- incomplete or insufficient answers,
- irrelevant answers,
- unclear or unhelpful clarification requests,
- false positive safety or content blocks,
- loss of relevant task context.

Set `is_trigger = FALSE` if the current assistant turn is adequate enough that it would not normally motivate repair, steering, or dissatisfaction, even if it is not perfect.

Explicit reaction rule

If explicit user reaction metadata is available and the reaction is thumbsDown, mark `is_trigger = TRUE` by default, unless there is strong contextual evidence that the reaction is not attributable to the current assistant turn.

A thumbsUp signal does not automatically require `is_trigger = FALSE`, but it is evidence against a trigger judgment unless the assistant turn clearly contains a user-relevant problem.

Retrieval-fallback rule

A retrieval or document-availability fallback is TRUE by default when the assistant states that no relevant documents, observations, or information are available without materially helping the user proceed.

Mark retrieval fallback as `is_trigger = TRUE` if the response:

- does not answer the user's actual request,
- gives no actionable next step,
- asks for documents or context only vaguely,
- is generic enough to fit many unrelated queries,
- or fails to provide substantive recovery support.

Mark retrieval fallback as `is_trigger = FALSE` if the response still provides meaningful recovery support, such as:

- a focused clarification question,
- a specific request for missing information,
- partial but useful guidance,
- a targeted reformulation suggestion,
- a precise pointer to relevant sources or workflow context,
- or safe general guidance that helps the user make progress.

Clarifying-question rule

A clarification question is not a trigger if it:

- shows partial understanding of the task, and
- asks for specific missing information needed to continue.

A clarification question is a trigger if it:

- is overly generic,
- shows little or no understanding,
- repeats prior confusion without progress,
- or does not help move the task forward.

Important note

`is_trigger` reflects a perceived user-relevant breakdown, not verified assistant error or root cause. It is judged primarily from the assistant turn itself and does not require the following user turn to contain an explicit repair label.

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7. Quick Decision Workflow

Use the following order for every annotation unit:

Step 1 — Read the current user query and assistant response together

Use prior context as needed.

Step 2 — Is the current assistant turn a trigger?

Annotate `is_trigger`.

Step 3 — Is the user continuing the same goal or starting a new one?

- same goal → likely Repair, Steering, or Task Progression
- new goal → T1

Step 4 — If same goal, what is the user mainly doing?

Ask:

- Are they correcting the assistant? → R5

- Are they asking for clarification because they do not understand? → R4
- Are they requesting more information to resolve a prior insufficiency or unresolved issue? → R3
- Are they adding or updating task information? → R2
- Are they repeating or rephrasing the same request without new information? → R1
- Are they constraining the output or terminology? → S1 / S2

Step 5 — Assign all applicable labels

Use multi-labeling only when more than one function is clearly present.

Step 6 — Choose one primary label

Use the priority rules in Section 11.

Step 7 — Add optional stance and end-state labels

Only if explicitly expressed.

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8. URS Families and Labels

F1 — Repair

Repair labels describe user behaviors that are commonly associated with resolving misunderstandings, ambiguities, insufficiencies, incorrectness, or other interactional difficulties. Some repair-labeled behaviors, particularly R3, may also occur during ordinary task progression and should be annotated based on observable user behavior rather than inferred assistant deficiency.

R1 — Repetition / Rephrasing

The user restates, rephrases or translates the same request without adding new task-relevant information.

Use when

- same goal,
- same content,
- no new parameter, constraint, or identifier.

Examples

- “What causes error F320?” → “Why does error F320 occur?”
- “How do I reset the drive?” → “How can I reset the drive?”
- “What causes error F320?” → “Warum tritt der Fehler F320 auf?”

Do not use when

- the user adds new details, conditions, or corrections → use R2
- the user explicitly says the assistant is wrong → use R5

R2 — Information Addition / Update

The user adds, updates, or narrows task-relevant information without explicitly framing the assistant as wrong.

Use when

- answering the assistant’s clarification question,
- adding context,
- adding identifiers, constraints, or conditions,
- updating user preferences or task parameters,
- self-correcting within the same user turn.

Examples

- “The machine is an S7-1500.”
- “It happens after 10 minutes.”
- “Actually, it is motor B.”
- “I need the German manual.”

Do not use when

- the user explicitly says the assistant’s prior answer was wrong → use R5

R3 — Request More Information

The user requests additional task-relevant information related to the current goal or topic.

This label captures requests for elaboration, confirmation, procedural guidance, resolution, or additional task-relevant information.

R3 is defined behaviorally rather than evaluatively. Assignment of R3 does not require a judgment that the preceding assistant response was incomplete, deficient, or problematic. Such requests may function either as repair following an insufficient response or as ordinary continuation within an ongoing task.

Use when

1. The user requests additional task-relevant information.
2. The user seeks elaboration, confirmation, procedural guidance, or resolution.
3. The request is prompted by information introduced in the preceding interaction.
4. The user appears to understand the assistant's meaning but seeks additional information related to the current task.

Do not use when

1. The user primarily does not understand the assistant's wording, meaning, rationale, or reference → **R4**.
2. The user contributes new information, constraints, observations, or corrections → **R2** or **R5**.
3. The user initiates a new goal or independent task → **T1**.

R4 — Clarification Request

The user asks for clarification because they do not understand the assistant’s meaning, wording, reasoning, or reference.

Use when

- the user does not understand a term,
- the user does not understand the rationale,
- the user does not understand what the assistant is referring to,
- the user does not understand how the answer should be interpreted.

Examples

- “What do you mean by calibration drift?”
- “Which parameter are you referring to?”
- “Why do you say it is a sensor issue?”
- “Can you explain that?”

Decision rule

- If the user understands but wants more to resolve an insufficiency → **R3**
- If the user does not understand → **R4**

R5 — Correction

The user explicitly indicates that the assistant's prior response was incorrect, misleading, or based on a misunderstanding, and provides corrected information.

Examples

- “No, the code is F320, not F302.”
- “That is not the conveyor motor, it is the feeder axis.”
- “I asked for the German manual, not the English one.”
- “No, according to the manual, F320 refers to encoder fault.”
- “That is incorrect; the log shows pressure loss, not overheating.”
- “The HMI says axis Z, not axis A.”

Use when

- the user states that the assistant is wrong or inaccurate;
- the user replaces information provided by the assistant with corrected information;
- the user corrects a misunderstood entity, parameter, condition, intent, language, assumption, or interpretation;
- the user provides evidence or supporting information while correcting the assistant.

Do not use when

- the user adds new information, constraints, observations, or preferences without explicitly correcting the assistant → **R2**
- the user requests clarification or additional information without indicating that the assistant was wrong → **R3**
- the user indicates misunderstanding of the assistant's wording or meaning without correcting factual content → **R4**

F2 — Steering

Steering labels capture user turns that shape how the assistant should respond, for example by constraining format, terminology, language, or output style.

S1 — Output Restriction

The user constrains the format, scope, verbosity, or structure of the answer.

Examples

- “Just give me the steps.”
- “Keep it short.”
- “Only list the spare parts.”
- “Answer in bullet points.”
- “Do not explain the theory.”

S2 — Language / Terminology Control

The user requests a specific language, terminology, naming convention, or level of technical language.

Examples

- “Answer in German.”
- “Use operator terms.”
- “Use the HMI labels.”
- “Do not use engineering jargon.”

F3 — Task Management

T1 — Task Change

The user starts a new task.

Use when

- the new request could stand alone as a separate goal,
- the previous goal is no longer being pursued.

Examples

- “Never mind. How do I reset the PLC?”
- “Forget the error code. I need the spare part number.”
- Previous turn: “How do I calibrate the sensor?” → Current turn: “What is error 123?”

Do not use when

- the user is still pursuing the same goal through a follow-up question.

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9. Multi-Label Policy

Assign all applicable `URS_labels` only when multiple functions are clearly present.

Choose exactly one `URS_primary`.

Examples

“No, it is F320. What should I do next?”
→ R5 + R3
Primary: R5

“Answer in German and keep it short.”
→ S2 + S1
Primary: choose the more central constraint in context

“That is wrong; the log shows pressure loss. Can you verify it?”
→ R5 + R3
Primary: R5

When in doubt, prefer the label that best captures the main communicative function of the turn.

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10. Primary Label Priority

If more than one URS label applies, choose the primary label using the following priority order:

1. T1
2. R5
3. R4
4. R3
5. R2
6. R1
7. S1
8. S2

Interpretation

- New goal initiation (T1) takes priority over within-goal behavior.
- Explicit correction outranks information addition or repetition.
- Clarification about misunderstanding outranks simple follow-up requests.
- Requests for additional information (R3) outrank information updates (R2) and repetitions (R1).

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11. Decision Rules

Rule A — Same goal or new goal?

- same goal → use R*, S*
- new goal → T1

Rule B — Clarification, repair-oriented info request, or task progression?

- user does not understand → R4
- user requests additional task-relevant information, elaboration, confirmation, procedural guidance, or resolution related to the current goal → R3
- user is simply continuing the task → none

Rule C — Correction or information addition?

- user explicitly says the assistant is wrong → R5
- user adds or updates information without explicit correction → R2

Rule D — Rephrase or update?

- same meaning, no new task information → R1
- added or revised task information → R2

Rule E — Requests for next steps?

- asks what to do next, how to proceed, or how to resolve the issue → R3
- merely acknowledges or authorizes a proposed next step → none

Rule F — Preference update or factual correction?

- preference or parameter change without explicit error claim → R2
- explicit correction of assistant output → R5

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12. Stance Labels

Annotate stance only if it is explicitly expressed.
Do not infer dissatisfaction from correction alone.

Label	Meaning
S+	positive
S−	dissatisfaction
Sd	doubt
Sf	frustration

Examples

- S+: “Perfect, thanks.”
- S−: “That is not helpful.”
- Sd: “Are you sure?”
- Sf: “This is the third time I have asked.”

Multiple stance labels may be assigned if clearly present.

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13. End-State Signals

Assign end-state labels only if success or abandonment is explicitly signaled in the current turn or immediate context.

Label	Meaning
E1	success
E2	abandonment

Examples

- E1: “That solved it.” / “Got it, thanks.”
- E2: “Never mind.” / “I will do it myself.”

Do not infer E2 from silence or lack of response unless your dataset explicitly marks abandonment.

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14. Edge Cases

14.1 Correction plus follow-up question

If the user both corrects the assistant and asks for more information:

- assign both labels if both are clearly present,
- choose R5 as primary depending on evidence.

Example:

- “No, it is F320. How do I fix it?”
    - R5 + R3 when the user both corrects the assistant and requests additional information.

14.2 Correction plus stance

Correction does not automatically imply negative stance.

Example:

- “No, it is F320.”
    - R5, no stance unless dissatisfaction is explicit

Example:

- “No, that is wrong and not helpful.”
    - R5 + S-

14.3 Same broad task, new subgoal

Use T1 only if the user starts a genuinely new goal.

Example:

- “What causes F320?” → “How do I reset the PLC?”
    - T1

Example:

- “Forget F320.” → “I need the spare part number for the encoder.”
    - T1

14.4 Generic assistant clarification

If the assistant asks a vague clarification question such as:

- “What do you mean?”
- “Can you clarify?”

without showing understanding or specifying what is missing, this may be a trigger.

If the user then rephrases or adds information, annotate the user turn normally (R1 or R2) and mark `is_trigger = TRUE`.

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15. Minimal-Pair Examples

R1 vs R2

“Why does F320 occur?” → “What causes F320?”
→ R1

“Why does F320 occur?” → “What causes F320 on an S7-1500 after startup?”
→ R2

R2 vs R5

“It happens after 10 minutes.”
→ R2

“No, it is motor B.”
→ R5

R3 vs R4

“What do you mean by drift compensation?”
→ R4

“Okay, but what should I do next exactly?”
→ R3

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16. Rationale Guidelines

Keep rationales short, concrete, and functional.

Include:

- the trigger cue if relevant,
- the main reason for the URS label,
- any necessary reference to context.

Good rationale examples

- “User rephrases same request after vague fallback → R1.”
- “User explicitly corrects assistant’s fault code → R5.”
- “User asks what assistant means by ‘calibration drift’ → R4.”
- “User starts a different goal after abandoning prior troubleshooting → T1.”
- “User requests next step after incomplete answer → R3.”